Reviewer(s)' Comments to Author:
Reviewer: 1

Comments to the Author
The manuscript presents an extensive database of 9601 Czech words rated on five dimensions: concreteness, imageability, AoA, valence and gender, by more than 4300 participants. The dataset is openly available, large in scope and using an original statistical approach. Overall, the resource is valuable and likely to be widely used in psycholinguistic research.

However, the manuscript requires substantial revisions before it can be accepted for publication.

We would like to thank the reviewer for their thoughtful and helpful comments on the work presented.

First of all, I find the beginning of the Introduction (p. 3-4) diffuse and hard to follow. The authors begin by referring a long history of research on the nature of word meaning, but not mentioning that theories of semantic representation are somehow fragmented and vary across multiple dimensions. For instance, one dimension is the extent to which theories invoke fixed lexical contents in forms of rules, features (e.g., Collins & Quillian, 1969) or prototypical examples (e.g., Rosch & Mervis, 1975) versus theories that propose reliance on inference and world knowledge (e.g., Elman, 2009). Another dimension is the extent to which meaning is captured through distributional co-occurrence patterns within language (e.g., Firth, 1968; Landauer & Dumais, 1997). This latter perspective refers to the idea that a word does not obtain its meaning in isolation, but rather from its linguistic context, which is basically the principle underlying the LLMs mentioned on page 4. In contrast to DSMs, other approaches ground language in non-linguistic representations (which is relevant for your dimensions of interest), etc. In general, I believe that the MS would benefit from some reorganization and from a deeper overview of theories that focus on word meaning.

We would like to thank the reviewer for raising this comment. We appreciate that there are a number of different theoretical perspectives on the nature of semantic representations and word meaning, with the reviewer highlighting work that has been well covered in numerous other papers that present similar norms. We originally wanted to keep the introduction much more focused on contemporary methodological issues, such as the need for more diverse language samples from human-derived data collection, and how these datasets may be used as benchmarks for computational approaches, such as the emerging trend of using LLMs. Given our main aims are heavily weighted towards an empirical and methodological contribution, we did not consider this specific theoretical background to be as central to our introduction.

On reflection however, our focus on methodological aspects could have come at a cost of not highlighting the wider theoretical work in as much detail. We have now reworked the introduction to give a fairer balance to the theoretical background, and to note the relevant work mentioned by the reviewer. We would like to highlight, that we have tried to prioritise breadth in the first few

pages over depth to ensure the scope is suitable, but we do provide more specific theoretical background on p.6-8, where we introduce each of the semantic dimensions individually, focussing on their relevance to both theoretical and empirical work. On p.3, we have added:

> *"Much of the work in this area has given a lot of attention to the nature of word meanings, but there has also been a range of theoretical perspectives that have tried to account for the way that semantics are represented, with much of the quantitative work on word meanings being closely tied to these theories. Some approaches assume that word meaning can be represented through relatively stable lexical content, such as semantic features or category structures (e.g., Collins & Quillian, 1969; Rosch & Mervis, 1975). Other accounts place greater emphasis on the role of inference, context and world knowledge in meaning construction (e.g., Elman, 2009). Distributional approaches, which propose that semantic representations emerge from patterns of linguistic co-occurrence, such that words derive meaning from the contexts in which they appear have also been influential (e.g., Firth, 1957; Landauer & Dumais, 1997). Finally, embodied perspectives place more emphasis on the role of perceptual, motor and affective experiences that help ground semantic knowledge (Barsalou, 1999; 2008). Normative datasets that aim to quantify semantic dimensions for the lexicon of a language, such as concreteness and valence, have acted as key resources allowing researchers to better test these theories and crucially, how to address issues related to how we understand semantic representation more broadly (see e.g., Davis & Yee, 2021; Wingfield & Connell, 2022)."*

Another content-related note concerns the choice of the selected dimensions. I agree that these are probably the most commonly investigated dimensions, although not exhaustive (i.e., arousal, familiarity, etc.). I also noticed that the "Relationships between variables" section (p.15-16) does not include specific predictions (e.g., "Although not underpinned by any specific theoretical motivation…", p. 15, l. 45), although the relationships among these dimensions of interest have been extensively investigated and documented (e.g., for relationship between AoA and imageability; see Cortese & Khanna, 2007; or AoA & affective measures, see Moors et al., 2013). The obtained predictions and results should therefore be interpreted more explicitly in relation to previous findings.

We thank the reviewer for bringing this to our attention. On their first point, we added additional text to highlight that the 5 dimensions being presented in the manuscript are not an exhaustive list of all possible semantic dimensions. On p.7 we have added:

> *"Below, we outline the theoretical and empirical background for each of the 5 dimensions that there is data for within the CzeLex norms. We focussed on these dimensions to ensure there is a multidimensional dataset available of distinct semantic norms for Czech, as was the case for other languages (e.g. Scott et al., 2019). Whilst there are a number of dimensions being presented in this paper. . Our main motivation*

for choosing this selection of dimensions was to provide researchers with a range of dimensions that have received considerable attention in the existing literature, i.e. AoA, imageability, concreteness and valence have all been studied in detail in other languages, but also include a more dimension – gender – which has received substantially less attention, as highlighted by Table 1. We acknowledge that this is not an exhaustive list and there are many other theoretically motivated dimensions of semantic representation (see Binder et al., 2016), that could be further investigated to better empirically understand semantic representations."

On the reviewer's second point, we stand by our decision not to provide a hypothesis test for these correlations, as the theoretical motivation for many of the relationships simply was not there. We believe this would mean reporting results where there were no clear theoretical arguments on which we can interpret the results. We certainly do acknowledge and note that several of the correlations have been reported previously, but agree that the links back to previous findings were sparse. We have updated the text on p.18 to more explicitly link back to relevant work, making direct comparisons to Scott et al's (2019) results.

"Given the number of correlations presented, we will not report on each individual relationship. However, there were a number of particularly strong linear relationships that can be observed. Concreteness and imageability had the strongest relationship within the five dimensions (r = .796), a pattern that has been well documented in previous studies; Scott et al., 2019, with Scott et al. reporting r = .91), whilst more modest relationships were found between gender and valence (r = .218), replicating the finding in Scott et al (2019) that more feminine words were evaluated more positively (with their study reporting r = -.42, although their gender scale was in the reverse direction to the present study). Likewise, there was also a negative relationship between AoA and imageability/concreteness (r = -.382 and -.216 respectively), with more imageable/concrete items being acquired earlier, which again was reported in Scott et al.'s (2019) analysis (with their study reporting r = -.49 and r = -.38 respectively). We do note that compared to Scott et al.'s results, the correlations reported here are evidently slightly smaller in size, this may be as a result of a much larger set of items that cover a more diverse lexical inventory. Nevertheless, the patterns are consistent across the English and Czech analyses."

There are also a couple of methodological issues that should be addressed. For instance, it is not clear what was the criteria for item selection (Stimuli section, p.9-10) and the composition of the stimuli with respect to parts of speech? This seems to be relevant because existing rating norms tend to favor nouns, resulting in more limited normative data (and potentially more limited understanding of lexico-semantic processing) of other PoS, such as verbs.

Thank you for raising this point. Our sample of words was derived predominately from the self-paced reading corpus of Czech Newspaper Headlines reported in Chromý et al. (2025) and a different norming study for Czech (Preininger et al., 2025), which covered a wide range of items used by Czech speakers, providing a good basis for an ecologically valid stimuli set, spanning a range of different PoS categories. As the distribution of words across those categories was

naturally uneven, we did carefully control for the distribution of words by PoS when creating the different word lists. We have updated the text on p.11 to reflect this methodological information and decision making.

In the Stimuli subsection:

> *"There were 1,745 adjectives, 5,713 nouns, 1,462 verbs and 681 items that had a different part of speech (such as adverbs, propositions, numerals, conjunctions), see Chromý et al., (2025) for details on the morphological annotation."*

In the Procedure subsection:

> *"Each list contained a range of words with different parts of speech, specifically we ensured that each list would contain 18/19 adjectives, 70/71 nouns, 20/21 verbs and 11/12 items that had a different part of speech. Participants rated a different list of words for each of the dimensions, with all participants being presented with all 5 of the dimensions during the experiment, but the lists were always randomly assigned within participants, so they never completed the same list of words across dimensions. There were 80 different lists of words in total."*

Also, I believe it is important to clarify how the rated stimuli were assigned across dimensions. That is, did participant rate each word on only one dimension, or could the same word appear across multiple dimensions? In my opinion, ratings the same item on multiple dimensions could lead to response biases (e.g., higher concreteness scores after rating the same stimuli in terms of imageability).

We agree that the clarity on this needed improving within the manuscript, and that this was also noted by R2. We have updated the text on p.11. Following on from the previous comment on PoS, we have also added the following to ensure readers understand that each dimension had a different list of words presented.

> *"Participants rated a different list of words for each of the dimensions, with all participants being presented with all 5 of the dimensions during the experiment, but the lists were always randomly assigned within participants, so they never completed the same list of words across dimensions. There were 80 different lists of words in total."*

Finally, based on the described procedure, the list of to-be-rated items for each dimension was presented simultaneously. How can the authors ensure that participants rated each word independently, that is, without referring to the remaining words in the list?

This is a good point, and we appreciate the reviewer drawing attention to this aspect of the procedure. We are aware that presenting items simultaneously may, in principle, allow participants to compare across words rather than evaluate each item independently. To reduce such effects, we randomised the order of the words for each participant, ensuring that each individual saw the items in a different sequence (with the exception of the calibrator word, which was always presented first). This procedure minimises systematic order effects across participants.

Moreover, given the relatively large number of items and the diversity of parts of speech included, it is unlikely that participants engaged in consistent cross-item comparisons that would meaningfully bias the ratings. Presenting items in this manner is also a common approach in norming studies (e.g., Scott et al., 2019), and is generally considered unlikely to substantially affect the reliability of the resulting ratings.

Minor concerns:

- P.10 and 15 "Error! Reference source not found.."; should it be a link to the OSF page

Our apologies for this, there was an error with the referencing to Tables in the source file, this has been corrected and the appropriate Tables are now referred to in the text. Also noted by R2.

- The third paragraph on p.3 is not complete (and in smaller font size), is it a part of the main text or a footnote?

Thank you for highlighting this, we believe this was an issue when exporting to pdf. It has now been resolved, and the paragraph should be more comprehensible now. Also noted by R2.

- I find the two paragraphs on LLMs on pages 4-5 somewhat out of place and more suitable for the Discussion (as the authors already partially do).

We appreciate the reviewer's comment here. When we considered the content of what should be included in the introduction, we felt that the emerging interest in LLMs for norming was a particularly pertinent topic to cover, not least with the wider interest in them for methodological approaches in psycholinguistics. As we wanted to highlight a very contemporary point of discussion in the literature, we did feel that it was relevant and we would prefer to keep it within the introduction as it adds a novel perspective that has not typically been covered in other norming studies. We felt it also helps contextualise the points made in the discussion, by providing readers with an overview earlier in the manuscript. Naturally, if the reviewer feels strongly we would be happy to revisit their comment.

- I am a little bit confused about the difference between calibrator and control items. Was there any? In the first paragraph of the Procedure section, you mentioned they were obtained based on the pilot data. Was that the only purpose of conducting a pilot and were data collected in the pilot study included in the final dataset?

We have reworded the text on p.12 to help readers understand the details of the calibrator and control words more clearly. We ran the pilot with a range of items to ensure that participants understood the instructions and there were no technical issues when hosting the experiment online using our servers. The items included the pilot were expected to have a range of ratings

based on previous datasets that have normed the dimensions in different languages, with the pilot results confirming this pattern. We didn't add much detail on the pilot to the updated manuscript as it felt a bit excessive, but we would be happy to if the reviewer thinks it would improve clarity.

> *"Within each list there was also a control word that was presented randomly with the other items. The calibrator words were included to ensure that participants were using the scale as intended at the very start of rating each dimension, whereas the control words were included primarily as a quality check within each dimension to ensure participants were rating the items in a suitable manner. See the data cleaning section for more details on this. Both the calibrator and control words were selected on the basis of them having reliable ratings with low variance in a pilot version of the experiment. See Table 2 for details of what the calibrator and control words were for each of the dimensions."*

- How did you decide about sample size?

We wanted to ensure that each item had a minimum of 10 ratings where participants did not choose the "I don't know the meaning of the word" option. This is in line with other large-scale studies, see e.g. Lynott et al. (2020), it was also motivated by Motamedi et al. (2019), who note that this is a suitable minimum threshold for rating iconicity. We appreciate that we are not presenting norms for iconicity or sensorimotor strength, but this is largely in line with other work in the norming literature. We have added additional details about the median N participants to Table 2 to highlight these descriptive data to supplement the information noted in Footnote 2 about the justification of a minimum sample size of 10.

- To assess the validity of your source, I suggest comparing your ratings for words shared with other Czech sources cited in Table 1 (e.g., AoA, gender, etc.).

Thanks for suggesting this. We ran a Pearson's correlation analysis on the data from the present study to the data reported in Preininger et al. (2025) for gender and valence, with the correlation coming out very strong, r = .99 for both. This result was largely expected, given the methods were identical for this specific comparison and the participant sample was drawn from the same university credit system. For the AoA correlations using the data from Łuniewska et al. (2019), we also observed a very strong relationship between the two datasets, r = .878, we would expect this to be slightly weaker given the much smaller number of items and the difference in sampling method. Finally, for the imageability data, we again observed a very strong relationship with r = .92. All of these correlations had p's < .001.

We have added this information to p.17:

> *"As an additional check of validity, we compared our ratings to those from previous studies on Czech using a Pearson's correlation analysis, specifically for AoA (Łuniewska et al., 2019), imageability (Smolík, 2019), gender and valence (Preininger et al., 2025),*

all correlations were found to be very strong (all having r's > .87 and p's < .001), indicating no major deviation from previous datasets in Czech."

- Figure 4; does nchar represent length of item? Should it be mentioned among "Additional lexical and processing variables"?

The nchar does represent orthographic length, we have added this to the suggested section on p.15.

"nchar: The orthographic length of the item. This was calculated as the string length in characters, ignoring any whitespace."

Thank you in advance for addressing my comments.

Reviewer 2:

Comments to the Author:

The manuscript describes a normative study including 9601 words of Czech language rated by 4626 native speakers on several lexical-semantic dimensions, namely Age of Acquisition, Concreteness, Gender, Imageability, and Valence, while also providing Zipf Frequencies, OLD20, and RT intercepts for the same set of words (RT intercepts being calculated from another study conducted by the same team). The data and the code are made freely available in accordance with the good scientific practices. In addition to providing significant amount of data for one underrepresented language, the manuscript also provides important methodological contributions in comparing three methods of estimating conglomerative measures from rating scale data. The authors conduct traditional analysis of calculating measures of central tendencies, in alignment with the typical approach in the field. However, they also apply more recent methods that take into account the ordinal nature of Likert scale data, as well as Bayesian statistics. While providing evidence on the alignment of the latter two approaches, they also engage in detailed analysis on the advantages they provide.

I find the research presented in the manuscript to be highly valuable for the field, and I specifically find the detailed comparison of different methods (including practical recommendations) precious for future studies in the field. Therefore, I recommend the manuscript for publication, with several comments/suggestions that I believe would benefit the quality of the manuscript.

We would like to thank the reviewer for their positive comments on the manuscript and the insightful comments they have provided to us. These have really helped us.

My one methodological concern is related to the use of random intercepts for words as estimates of reaction latencies. Although I have been thinking about using them myself, I was confronted with some counterarguments. For example, Kliegl et al (2011) advise against such practices (for BLUPS, but I believe it generalizes to random intercepts, as well). I have to be honest and say that I am still contemplating on the debate myself, but regardless, I believe that more attention should be devoted to the matter and more elaboration should be included in the manuscript regarding this issue.

Kliegl R, Wei P, Dambacher M, Yan M and Zhou X (2011) Experimental Effects and Individual Differences in Linear Mixed Models: Estimating the Relationship between Spatial, Object, and Attraction Effects in Visual Attention. Front. Psychology 1:238. doi: 10.3389/fpsyg.2010.00238
Houslay, T. M., & Wilson, A. J. (2017). Avoiding the misuse of BLUP in behavioural ecology. Behavioral ecology: official journal of the International Society for Behavioral Ecology, 28(4), 948–952. https://eur01.safelinks.protection.outlook.com/?url=https%3A%2F%2Fdoi.org%2F10.1093%2Fbeheco%2Farx023&data=05%7C02%7Cjames.brand%40edgehill.ac.uk%7C7104cca5bfbd423c147808debf34fd06%7C093586914d8e491caa760a5cbd5ba734%7C0%7C0%7C639158431480395030%7CUnknown%7CTWFpbGZsb3d8eyJFbXB0eU1hcGkiOnRydWUsIlYiOiIwLjAuMDAwMCIsIlAiOiJXaW4zMiIsIkFOIjoiTWFpbCIsIldUIjoyfQ%3D%3D%7C0%7C%7C%7C&sdata=91Y

49ATruA8%2FJmm%2F5X6hk2E60xMPs3ClE22s7B9ivXg%3D&reserved=0

We really appreciate your thoughts on this, and value your honesty when it comes to considering what approach to use, given that it is still an unclear picture in the literature and many other researchers may also be grappling with this dilemma. The original motivation to use the random intercepts stems from much of the thinking about how to calculate the normative estimates, i.e. the importance of accounting for by participant variance, something that a descriptive statistic such as the mean would not be able to account for.

Nevertheless, we completely agree with the reviewer that the original attention given to this in the manuscript was quite limited. Our primary reason for this was that the source of these estimates was not a key aim we wanted to focus on within the manuscript, as they were more intended to provide the validation of the norms and act simply as a measure of lexical processing. Indeed, we are aiming to do a much deeper analysis on the issues discussed in this topic within the cited work that is currently in prep by Brand et al., where we will have more scope to address the differences in calculating aggregate estimates using different approaches.

As the manuscript is already quite lengthy, we would prefer to avoid a further detailed addition on this topic and would instead prefer to address this in a more condensed manner, specifically by offering readers insights into the difference between the models using the intercepts and the traditional mean, for which there are no clear differences when interpreting the patterns of results being reported in this section of the analysis. This similarity in results may be explained by the simplicity of the modelling procedure, as we are not extracting the intercepts from a model that contained any fixed effects, as this was not theoretically or empirically of interest, unlike in the Kliegl et al. (2011) paper the reviewer cites.

The text has been updated on p.23:

> *"For reference, as we were using measures for RTs and accuracy derived from mixed-effects modelling, instead of the traditional descriptive statistic of the mean, we conducted parallel analyses of all the results reported below and found no indication that the two approaches led to any differences in the pattern of results, indeed they were almost identical in terms of the general patterns being reported. Whilst there are mixed views on the suitableness of using both the random intercepts (see Houslay & Wilson, 2017; Kliegl et al., 2011) and the mean (Baayen et al., 2008; Drager & Hay, 2012), we report below only the analysis from the random intercepts approach as we feel it provides more control over participant-level variance in a lexical decision task, which although is outside the scope of the present paper, is explored further in Brand et al (in prep)."*

We have also added some additional text on p.15, to highlight the lack of main effects included in the model:

> *"Using the raw response data for reaction times (RTs) where participants responded accurately to the item, we ran a linear mixed-effects model, with a global intercept and random intercepts for participant and item, meaning there were no fixed effects included as we did not wish to control for any predictor variables."*

My other concerns are stylistic by nature. My dominant concern is the repetitiveness of the introduction, in relation to the multiple attempts to stress the importance and significance of the collected data. On multiple occasions they stress that they collected large amount of data for important semantic dimensions in understudied language. I believe it would be enough to mention this in the abstract, and later in the discussion. Along the same line, the introduction (in addition to the mentioned repetitiveness of the significance) is more technical than substantial, listing the dimensions in question and pointing to other sources for a deeper understanding of their significance. My advice would be to avoid the repetitiveness and rewrite the introduction to explain in more detail why these dimensions are relevant for the psychology of language. I am not suggesting writing a long theoretical introduction, but to merely broaden to some extent the description of the dimensions in question (with a focus on their psychological importance, relating them to models in more detail).

We thank the reviewer for this helpful and constructive comment, and we note that a similar concern was also raised by Reviewer 1. In response, we have substantially revised the Introduction to provide greater theoretical breadth and to more clearly articulate the psychological relevance of the target dimensions, including more explicit links to models of semantic representation. We hope that these revisions strengthen the conceptual framing of the paper. For further details, we refer the reviewer to our response to Reviewer 1's first comment, where these changes are described in more detail.

Minor points:

Abstract: I would suggest including some of the insights obtained in the study in the abstract.

We agree with the reviewer here, and have updated the abstract to include more insights from the analysis.

> *"We demonstrate that the dataset has good reliability and that specific dimensions predict measures of lexical processing in a lexical decision task, irrespective of the aggregation method used."*

Page 3, Lines 46-58: There is a paragraph in Italic letters that seems to be interrupted.

Thanks for highlighting this, R1 has also noted this and the paragraph has been updated to be as originally intended.

Page 10, line 34: There seems to be a corrupted link for the reference in this line.

Also noted by R1 and the text has been updated to correct the error.

Page 10, Procedure: Please include more details to explain more clearly what was presented to each participant. From the current description it is not fully clear how many dimensions and how

many different words each participant was presented with. For example, did one participant rate the same set of words on multiple scales, or were the words counterbalanced etc.

Thank you for this comment, we agree that the level of detail could be improved and this also aligns with R1's comment on part of speech distributions. We direct the reviewer to the response provided to R1.

Page 13, line 40: The authors mention "psychological coding", but it is not clear what they refer to. Please explain in more detail.

This was an error on the author's part and was meant to read 'phonological coding'. The text has been updated.

Page 14, line 3: There is a sentence that seems uninterpretable: "We then extracted the random intercepts for the item variable, providing goes with estimates of processing." Please rephrase.

This was another error by the author, the text "providing goes with estimates of processing' has been deleted. We would like to thank the author again for their sharp eye on these.

Page 14, line 60: The authors mention Brysbaert, 2019, but the reference is not listed.

We have added the following reference to the references section to address this:

Brysbaert, M. (2019). How many participants do we have to include in properly powered experiments? A tutorial of power analysis with reference tables. *Journal of Cognition*, 2(1), 16.

Page 15, line 22: There seems to be a corrupted link for the reference in this line.

This has been updated to reference Table 2 now.

Page 15, lines 54-56: The authors declare that they intentionally did not present statistical significance for the correlation coefficients, aiming at pure description. However, I believe that information of statistical significance would be useful, nevertheless.

We would argue against doing this, we did not have any specific hypotheses and believe that reporting significance values from null-hypothesis testing would not be a suitable option as there we did not have a research/alternative hypothesis, alongside no motivation that is grounded in theory. We believe that providing the correlation coefficients does provide a useful insight into the relationships being analysed, without the need for significance values that would warrant a theoretically motivated explanation, which was always our intention. We would prefer to maintain a descriptive approach to this analysis, as there was no underlying theoretical motivation for all of the comparisons being made. We note that this was also mentioned by R1, and provide additional details on how we have added more information to the manuscript to make comparisons to previously reported correlations between the variables in the literature.